Modified Equations for Symplectic Methods applied to Hamiltonian Systems

(GNI sections IX.3, IX.9)

Recall that when we apply a RK method to

$$u'(t) = f(u)$$

the numerical solution can be viewed as the exact solution of a modified equation

$$\tilde{u}'(t) = f(\tilde{u}) + h f_2(\tilde{u}) + h^2 f_3(\tilde{u}) + \cdots$$

Previously we expressed this as a B-series.

Thm. (GNI Thm. IX.3.1)

Consider a symplectic method applied to a Hamiltonian system

$$u'(t) = J^{-1} \nabla H(u)$$

Then the modified equation is also Hamiltonian and has the form

$$J \tilde{u}'(t) = \nabla H(\tilde{u}) + h \nabla H_2(\tilde{u}) + h^2 \nabla H_3(\tilde{u}) + \cdots$$

Proof: This is by induction on j for H_j. For $j=1$ it's clearly true since the original problem is Hamiltonian. Assume

$$f_j(u) = J^{-1} \nabla H_j(u) \quad \text{for } j=1,2,\dots,r$$

We want to show that there exists H_{r+1} s.t.

$$f_{r+1}(u) = J^{-1} \nabla H_{r+1}.$$

We will use GNI Lemma VI.2.7:

If the Jacobian $f'(u)$ is symmetric then there exists $H(u)$ such that

$$f(u) = \nabla H(u)$$

Proof:

$$H(u) = \int_0^1 u^T f(tu)\,dt$$

Compute $\nabla H(u) = f(u)$.

Let Φ_h denote the solution operator of the ME

$$u'(t) = f(u) + hf_2(u) + h^2 f_3(u) + \cdots$$ and

let $\phi_{r,h}$ denote the solution operator of the truncated ME

$$u'(t) = f(u) + hf_2(u) + \cdots + h^{r-1} f_r(u).$$

which is also a Hamiltonian system.

We have

$$\Phi_h(u_0) = \phi_{r,h}(u_0) + h^{r+1} f_{r+1}(u_0) + \mathcal{O}(h^{r+2})$$

and (differentiating w.r.t. u_0)

$$\Phi'(u_0) = \phi'_{r,h}(u_0) + h^{r+1} f'_{r+1}(u_0) + \mathcal{O}(h^{r+2}) \qquad (1)$$

Since the problem is Hamiltonian and the method symplectic, Φ_h is symplectic:

$$(\Phi_h')^T J \Phi_h' = J$$

So, using (1) and $\phi'_{r,h} = I + \mathcal{O}(h)$

$$J = (\Phi_h')^T J \Phi_h' =$$

$$(\phi_{r,h}'(u_0) + h^{r+1} f_{r+1}'(u_0))^T J (\phi_{r,h}'(u_0) + h^{r+1} f_{r+1}'(u_0)) + \mathcal{O}(h^{r+2})$$

$$= \underbrace{\phi_{r,h}'(u_0)^T J \phi_{r,h}'(u_0)}_{J} + h^{r+1}\Big(f_{r+1}'(u_0)^T J \cancel{\phi_{r,h}'(u_0)}$$

$$+ \cancel{\phi_{r,h}'(u_0)^T} J f_{r+1}'(u_0)\Big) + \mathcal{O}(h^{r+2})$$

$$J = J + h^{r+1}\underbrace{\left(f_{r+1}'(u_0)^T J + J f_{r+1}'(u_0)\right)}_{=0} + \underbrace{\mathcal{O}(h^{r+2})}_{J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}}$$

$$f_{r+1}'(u_0)^T J = -J f_{r+1}'(u_0) = J^T f_{r+1}'(u_0) = (f_{r+1}'^T(u_0) J)^T$$

So $J f_{r+1}'(u_0)$ is symmetric,

So $J f_{r+1}'(u_0)$ is symmetric. Therefore, there exists H_{r+1} s.t. (See GNI Lemma VI.2.7.)

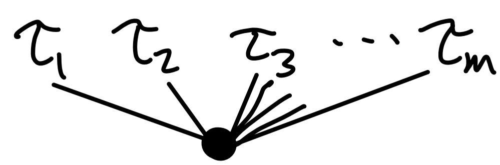
$$f_{r+1}(u) = J^{-1} \nabla H_{r+1}(u).$$

Further Results

In GNI IX.9.2-3, the functions $H_j(u)$ are constructed explicitly. They are linear combinations of "elementary Hamiltonians".

We write

$$\tau = [\tau_1, \tau_2, \dots, \tau_m]$$



Then

$$H_\bullet(u) = H(u)$$

$$H_\tau(u) = H^{(m)}(u)\big(F(\tau_1)(u), \dots, F(\tau_m)(u)\big)$$

where $F(\tau_j)(u)$ is an elementary differential.

It turns out that many of these terms cancel out, so only a subset of trees need to be considered in practice.

For every symplectic method, a modified Hamiltonian is conserved. But no symplectic method conserves the true Hamiltonian for all Hamiltonian systems

(except the "exact solution" method).

Homework (due after Eid)

1. For the implicit midpoint rule applied to the pendulum, find the functions H_2, H_3 such that the modified equation is

$$u'(t) = J^{-1}\nabla\left(H(u) + hH_2(u) + h^2H_3(u)\right) + \mathcal{O}(h^3)$$

2. GNI Chapter IX problem 10.

3. Continuation of the previous homework: apply a non-symplectic RK method of your choice to either the pendulum or the nonlinear oscillator. Compare the error (as a function of time, over long times) obtained with the non-conservative method

versus that obtained when using projection or relaxation.